



Supplementary Information for

Massive parallel variant characterization identifies *NUDT15* alleles associated with thiopurine toxicity

Chase C. Suiter^a, Takaya Moriyama^a, Kenneth A. Matreyek^b, Wentao Yang^a, Emma Rose Scaletti^{c,d}, Rina Nishij^a, Wenjian Yang^a, Keito Hoshitsuki^a, Minu Singh^e, Amita Trehan^e, Chris Parish^a, Colton Smith^a, Lie Li^a, Deepa Bhojwani^f, Liz YP Yuen^g, Chi-kong Li^h, Chak-ho Liⁱ, Yung-li Yang^j, Gareth J Walker^{k,l}, James R Goodhand^{k,l}, Nicholas A Kennedy^{k,l}, Federico Antillon Klusmann^m, Smita Bhatiaⁿ, Mary V. Relling^a, Motohiro Kato^o, Hiroki Hori^p, Prateek Bhatia^e, Tariq Ahmad^{k,l}, Allen E. J. Yoeh^{q,r}, Pål Stenmark^{c,d}, Douglas M. Fowler^{b,s,t}, and Jun J. Yang^a

^aDepartment of Pharmaceutical Sciences, St. Jude Children's Research Hospital, Memphis, TN, USA; ^bDepartment of Genome Sciences, University of Washington, Seattle, WA, USA; ^cDepartment of Biochemistry and Biophysics, Arrhenius Laboratories for Natural Sciences, Stockholm University, Sweden; ^dDepartment of Experimental Medical Science, Lund University, Lund, Sweden; ^eDepartment of Pediatrics, Advanced Pediatrics Centre, PGIMER, Chandigarh, India; ^fDepartment of Pediatrics, Children's Hospital of Los Angeles, Los Angeles, California, USA; ^gDepartment of Pathology, Hong Kong Children's Hospital, Hong Kong; ^hDepartment of Paediatrics, The Chinese University of Hong Kong, Hong Kong; ⁱDepartment of Paediatrics and Adolescent Medicine, Tuen Mun Hospital, Hong Kong; ^jDepartment of Laboratory Medicine and Pediatrics, National Taiwan University Hospital, College of Medicine, National Taiwan University, Taipei, Taiwan; ^kDepartment of Gastroenterology, Royal Devon and Exeter Hospital NHS Foundation Trust, Exeter, England; ^lIBD Pharmacogenetics Group, University of Exeter, Exeter, England; ^mUnidad Nacional de Oncología Pediátrica, Guatemala City, Guatemala, Francisco Marroquin Medical School, Guatemala City, Guatemala; ⁿDivision of Pediatric Hematology/Oncology, Institute for Cancer Outcomes and Survivorship, School of Medicine, University of Alabama at Birmingham; ^oDepartment of Pediatric Hematology and Oncology Research, National Center for Child Health and Development, Tokyo, Japan; ^pDepartment of Pediatrics, Mie University Graduate School of Medicine, Mie, Japan; ^qCentre for Translational Research in Acute Leukaemia, Department of Paediatrics, Yong Loo Lin School of Medicine, National University of Singapore, Singapore; ^rCancer Science Institute of Singapore, National University of Singapore, Singapore; ^sDepartment of Bioengineering, University of Washington, Seattle, WA, USA; ^tGenetic Networks Program, CIFAR, Toronto, ON, Canada

Jun J. Yang, PhD
Department of Pharmaceutical Sciences
St. Jude Children's Research Hospital, MS313
262 Danny Thomas Place Memphis, Tennessee 38105-3678
Email: jun.yang@stjude.org
Phone: 901-595-2517

This PDF file includes:

Supplementary text
Figures S1 to S7
Tables S1 to S5
Legends for Datasets S1
SI References

Other supplementary materials for this manuscript include the following:

Datasets S1

Supplementary Information Text

Intracellular abundance of the Arg139Cys variant in the landing pad HEK293T cells

Wildtype (WT) *NUDT15-EGFP* was cloned into attB-IRES-mCherry backbone by using NEBuilder HiFi DNA Assembly Cloning Kit (New England BioLabs) according to the manufacture's instruction. *NUDT15* variant c.C415T (encodes Arg139Cys) was generated via site-directed mutagenesis by using QuikChange Lightning Site-Directed Mutagenesis Kit (Agilent Technologies). The vectors were transfected into the landing pad HEK293T cells for recombination at the *AAVS1* locus, and *NUDT15* expression was induced using 2 µg/mL doxycycline(1, 2). Recombinant cells expressing WT or Arg139Cys *NUDT15* were identified by flow cytometry as BFP-negative and mCherry-positive. *NUDT15* protein abundance in these cells was quantified as EGFP/mCherry ratio (**SI Appendix Fig. S1A**).

NUDT15 mutagenesis library and the landing pad cellular model

Library construction, cloning, and transfection were performed according to our previously published procedures, with slight modifications(1, 2). First, a *NUDT15* site-saturation mutagenesis oligo library with flanking attB sequences was purchased from Twist Biosciences (Menlo Park, CA). The oligo library was introduced into SacI and SphI site on the attP-IRES-mCherry backbone by NEBuilder (New England Biolab), with EGFP coding sequenced fused to the C-terminus of *NUDT15*. An 18-bp barcode was inserted between the stop codon of EGFP and the beginning of the IRES sequence, using T4 ligase. The relationship between variants and barcodes was established by single molecule real time (SMRT) sequencing. The Circular Consensus Sequencing 2 (version 2.0.0, with default parameters) was first used to measure sequencing error and detect consensus sequences of *NUDT15* coding region, using base call files of SMRT sequencing (<https://github.com/PacificBiosciences/unanimity/blob/master/doc/PBCCS.md>). Consensus sequences were then aligned to *NUDT15* reference sequence (NM_018283) by Burrows-Wheeler Aligner(3). Barcodes and their associated *NUDT15* consensus sequences were extracted by scripts from Matreyek et. al(1).

The barcoded *NUDT15* library was integrated into the *AAVS1* locus in the landing pad HEK293T cell line. Cells were transfected with a Bxb1 expression plasmid and the *NUDT15* library on

successive days. Cells were cultured for 48 hours to allow for recombination, then *NUDT15* expression was induced using 2 µg/mL doxycycline. Successfully recombined cells were identified by flow cytometry as BFP-negative and mCherry-positive.

Protein abundance-based screen of *NUDT15* variants

Recombinant cells were binned into four equally populated groups based on their EGFP/mCherry ratio, and genomic DNA was separately isolated for each bin of cells. PCR was performed to amplify barcode regions and to add a unique molecular index. Paired-end 100bp sequencing was performed on Illumina HiSeq 2000. Sequence analyses (reads mapping and alignment, variant calling) were performed in the accordance with the previous study(1). Sequenced barcodes counts were then assigned to each *NUDT15* variant according to results from SMRT sequencing generated as above. A weighted average value a_v for each variant (i.e., weighted variant frequency) was calculated on the basis of its read counts in each bin, as described previously(1).

The abundance score $\hat{a}_v$ was obtained by normalizing a_v from 0 to 1 using the formula below

$$\hat{a}_v = \frac{a_v - \bar{a}_l}{a_{wt} - \bar{a}_l} \quad a_l \in a_{q=0.05}$$

where $\bar{a}_l$ and a_{wt} are the mean value of variants in the lowest 5th percentile and for WT, respectively.

Thiopurine cytotoxicity-based screen of *NUDT15* variants

Fold change (s) in variant frequency before and after thiopurine treatment was used to estimate “drug sensitivity score” for each variant (**SI Appendix Dataset S1**) using a procedure similar to that for abundance score:

$$\hat{s}_v = \frac{s_v - \bar{s}_l}{s_{wt} - \bar{s}_l} \quad s_l \in s_{q=0.05}$$

where $\bar{s}_l$ and s_{wt} are mean values for variants within the lowest 5th percentile and WT, respectively.

We also performed experiments to determine whether *NUDT15* variants affected HEK293T cell growth. *NUDT15* library-transfected cells were grown for 2 or 6 days at which time genomic DNA was extracted for massive parallel sequencing. Variant count in each cell pool was enumerated to determine whether a variant promoted or retarded cell growth during this period. Of 2,845 evaluable

variants, we identified four (0.14%) with a significant effect of proliferation with a range of abundance, drug sensitivity, and NUDT15 activity scores (**SI Appendix Table S5**).

Bioinformatic prediction of *NUDT15* variant function

Potential damaging effects of *NUDT15* variants were predicted using a variety of bioinformatic algorithms: 837, 837, 837 and 834 variants were included for CADD (v1.4)(4), REVEL(5), PolyPhen2(6) and SIFT(7) prediction, respectively (**SI Appendix Dataset S1**). The association of NUDT15 activity score with each bioinformatic prediction was analyzed by using the Spearman (CADD and REVEL) or the Rank Sum test (PolyPhen2 and SIFT), respectively.

Orthogonal validation of *NUDT15* variant function

Fourteen *NUDT15* variants were selected to validate the abundance-based screen using orthogonal assays. First, individual *NUDT15* variants were introduced to the attB-NUDT15-GFP-IRES-mCherry by site directed mutagenesis using an Agilent QuikChange kit. Each variant was individually introduced into HEK293T cells by transfection, and the EGFP/mCherry ratio was quantified by flow cytometry to indicate intracellular NUDT15 protein levels.

In a second assay, variant *NUDT15* was cloned into the pColdII expression vector with an N-terminal His tag via site directed mutagenesis. Variant NUDT15 protein was expressed in *E. Coli*, purified to homogeneity, and then subjected to thermal stability assay using Sypro®Orange (Molecular Probes) to determine T_m value (melting temperature), as described previously(8).

The correlation of individually assessed EGFP/mCherry ratio or T_m values with abundance score of these 14 variants were statistically evaluated by using the Spearman test. The GFP/mCherry ratio and T_m values were calculated from three independent measurements.

As an orthogonal validation of the drug sensitivity score, we selected nine variants to individually test their impact on thiopurine metabolism. Variants were introduced individually into HEK293T cells as described above. Cells were then incubated with 3 μ M TG for 24hr, after which genomic DNA was extracted to measure thioguanine incorporation. DNA-TG quantification was performed using the LC-MS method following procedures established previously by us(8). The correlation of DNA-TG for each variant with their drug sensitivity score was evaluated using the Spearman test.

Association of *NUDT15* variants with thiopurine toxicity in patients

Considering the differences in how thiopurine therapy is given in different diseases, drug toxicity was evaluated based on tolerated dosage (in patients with acute lymphoblastic leukemia [ALL]) or the degree of myelosuppression at a given dosage (in patients with inflammatory bowel diseases [IBD]), as we described previously(8-13). For patients with ALL, MP dosage was clinically adjusted during the maintenance phase according to a pre-defined range of white blood cell (WBC) count and/or neutrophil count (ANC), and tolerated MP dosage was defined as the average of the daily dosage over at least 14 days after appropriate dose titration. Standard MP dosage and target WBC/ANC account were comparable with slight variation across different ALL treatment protocols included in this study (**SI Appendix Table S3**). For patients with IBD, thiopurine hematopoietic toxicity was evaluated using a multi-step algorithm as described previously(9), with myelosuppression defined as $WBC \leq 2.5 \times 10^9$ cells/l during thiopurine therapy without dose adjustment.

NUDT15 variants were divided into “rare” or “common”: variants with an allele frequency > 1% in at least one population in the gnomAD cohort (<https://gnomad.broadinstitute.org/>) are considered as common (Arg139Cys, Arg139His, and Val18Ile), whilst others variants are classified as rare (Gln6Glu, Arg11Gln, Lys33Glu, Arg34Thr, Val75Gly, Ser83Tyr, and Val93Ile). For common variants, we statistically tested the association of genotype with tolerated MP dosage during ALL therapy. Thus, tolerated MP dosage from patients carrying the *NUDT15* variant was compared to those with WT genotype (matched on race or ethnicity), with P value estimated using the linear regression model after adjusting for cohort.

For rare variants, five were identified in patients with ALL (Gln6Glu, Arg11Gln, Ser83Tyr, Val93Ile, and Arg34Thr) and two from the IBD cohort (Lys33Glu and Val75Gly). ALL patients carrying Gln6Glu, Arg11Gln, Ser83Tyr or Val93Ile tolerated MP dosage of 58.3, 50.0, 52.9 or 50.5 mg/m², all of which were within 10% of the average dosage of ancestry-match WT controls from each protocol(8, 10, 12, 13). By contrast, the MP dosage of two subjects with the rare variant Arg34Thr was 17.9 or 16.4 mg/m², representing 40.3 % and 36.4 % of WT controls, thus classified as “toxicity-related”. Two rare variants identified in the IBD patients (Lys33Glu and Val75Gly) were both

classified as “toxicity-related” on the basis of myelosuppression during thiopurine therapy as described above(9).

To determine whether the prevalence of individuals at risk of thiopurine toxicity due to NUDT15 deficiency is higher than what would be expected by chance, we compared the observed frequency (63%) based on NUDT15 activity score to the frequency of thiopurine toxicity in multi-ethnic population treated with this drug (9%)(10). The statistical significance was evaluated using Fisher exact test.

The full result of the association analysis is provided in **SI Appendix Table S1**.

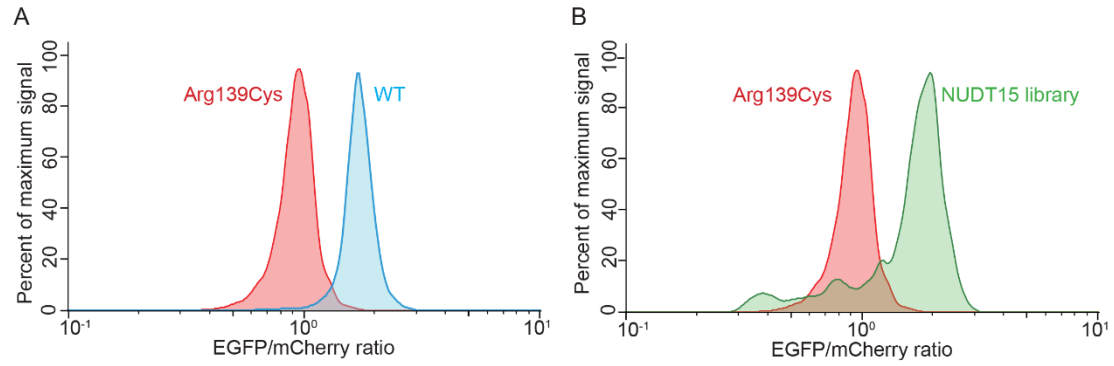


Fig. S1. Flow cytometry analysis of HEK293T cells expressing different NUDT15 variants. A) Cells were transfected individually with either wildtype (WT) NUDT15 or the Arg139Cys variant (fused with EGFP), and fluorescence was measured by flow cytometry and normalized to mCherry to account for differences in transfection efficiency. **B)** Similarly, fluorescence was measured for cells transfected with the Arg139Cys variant or the entire *NUDT15* variant library.

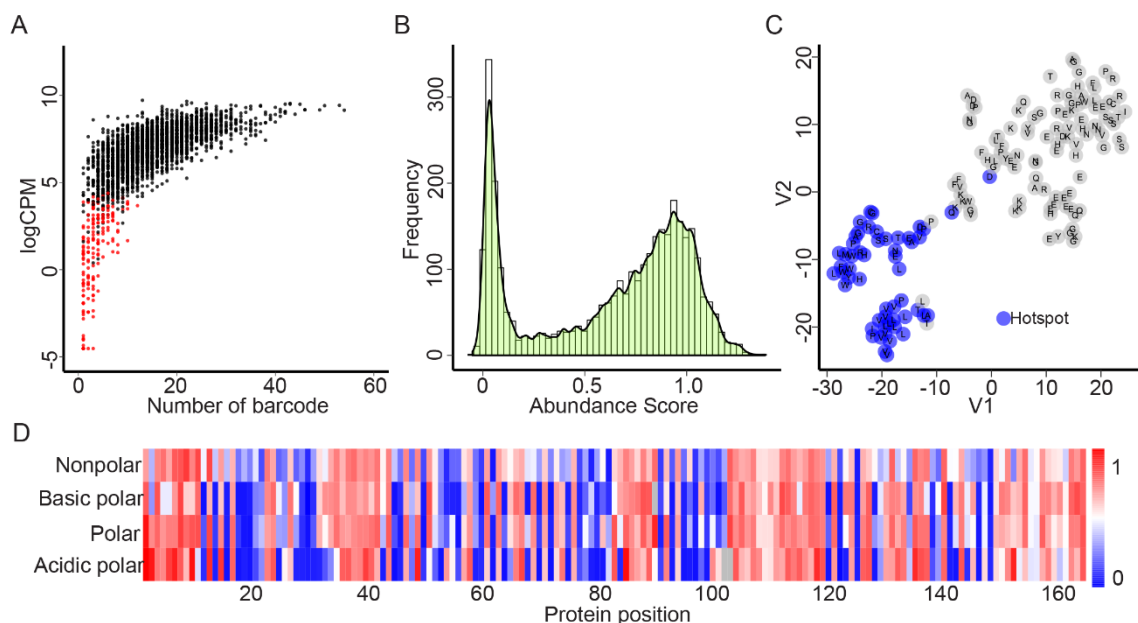


Fig. S2. Protein abundance-based functional screen of *NUDT15* variants. **A)** Correlation between sequencing depth and the number of barcodes across 3,097 *NUDT15* variants, with each dot representing a unique variant. After quality control, 154 variants (red dots) were removed and 2,922 variants (black dots) remained for further analysis. **B)** Distribution of variant abundance score. **C)** t-SNE analysis identified residues at which genetic variation has similar effects on *NUDT15* protein stability. Each dot represents a unique residue and clustering was on the basis of similarity and differences in the abundance score of 19 variants at each residue. Blue highlight indicates hotspot residues (see Methods). All amino acids listed are for WT *NUDT15*. **D)** Heatmap of abundance score at each position, summarized by variant amino acid's polarity. Abbreviation. CPM, count per million.

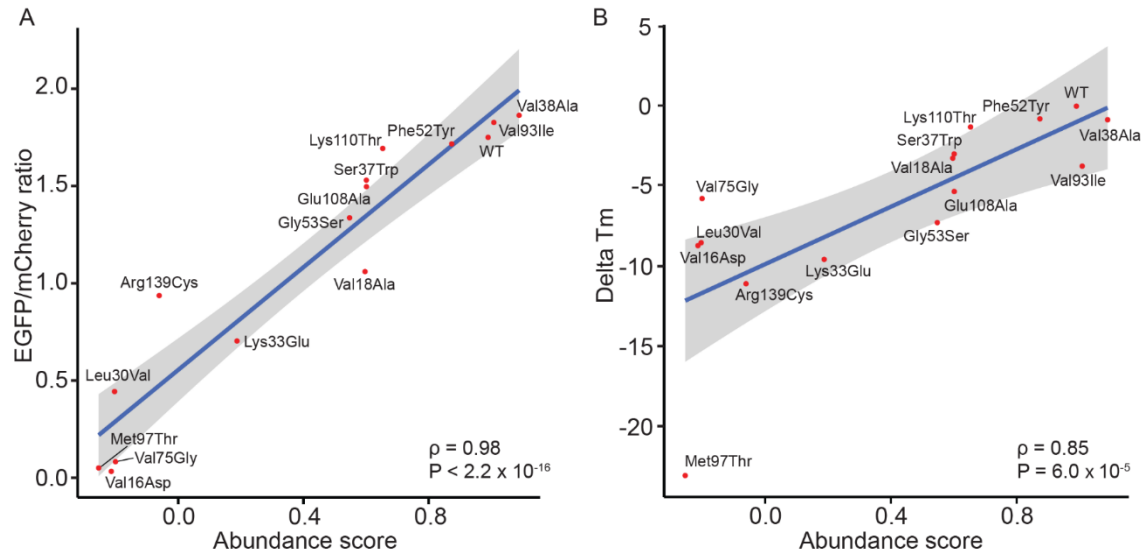


Fig. S3. Orthogonal validation of *NUDT15* variant abundance score. 14 *NUDT15* variants with a wide range of abundance scores were selected for validation. **A)** Fluorescence was measured in cells transfected with each *NUDT15* variant individually and plotted against estimated abundance score. **B)** Each of the 14 variant protein was expressed in *E. coli*, purified to homogeneity, and subjected to thermostability assay. Melting temperature was determined using the Sypro®Orange and correlated with the abundance score calculated from the high throughput screen. P values were estimated using the Spearman test.

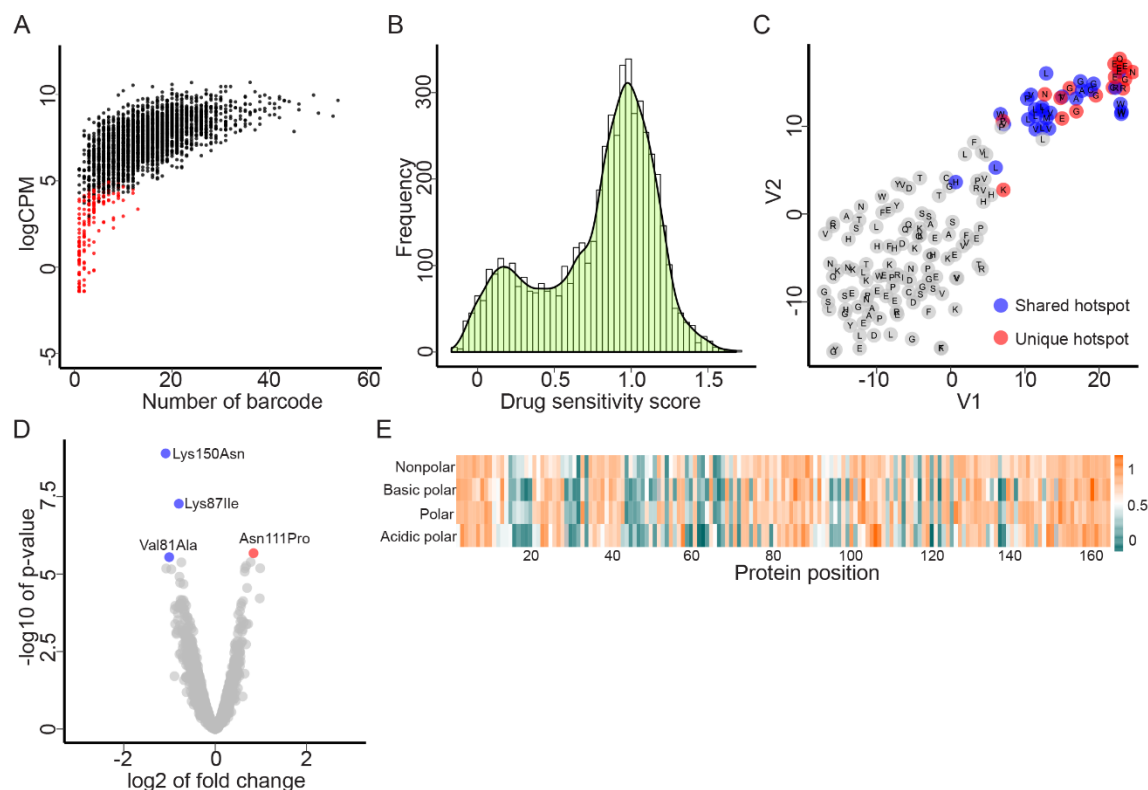


Fig. S4. Drug sensitivity-based functional screen of *NUDT15* variants. **A)** Correlation between sequencing depth and the number of barcodes across 3,097 *NUDT15* variants, with each dot representing a unique variant. After quality control, 142 variants (red dots) were removed and 2,935 variants (black dots) remained for further analysis. **B)** Distribution of variant drug sensitivity score. **C)** t-SNE analysis identified residues at which genetic variation has similar effects on thiopurine cytotoxicity *in vitro*. Each dot represents a unique residue and clustering was on the basis of similarity and differences in the drug sensitivity score of 19 variants at each residue. Blue and red dots indicate residues as hotspots shared with abundance score-based assay or independently detected by drug screen-based assay, respectively. see Methods). All amino acids listed are WT. **D)** Volcano plot describes changes in variant frequency between day 6 and day 2 in HEK293T cells without thiopurine, as a measurement of a variant's effect on cell growth. Red and blue points represent those show significantly positive or negative change (adjusted p-value <0.01), respectively. **E)** Heatmap of drug sensitivity score at each position, summarized by variant amino acid's polarity. Abbreviation. CPM, count per million.

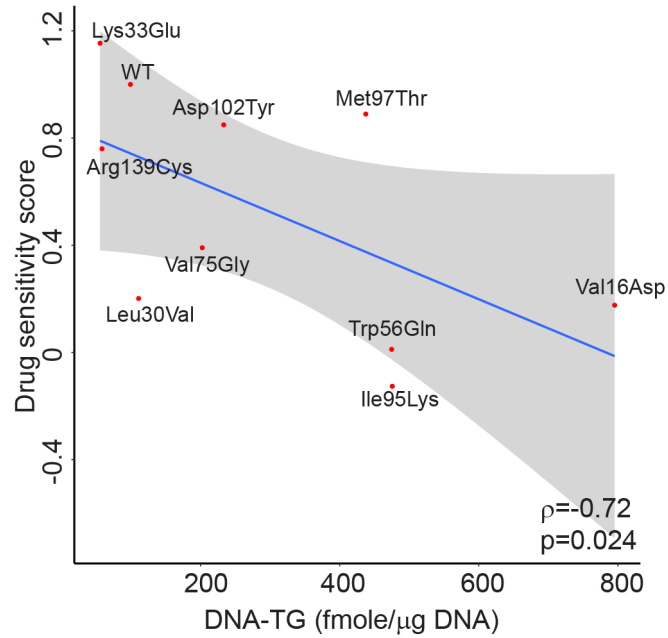


Fig. S5. Orthogonal validation of *NUDT15* variant drug sensitivity score. Nine *NUDT15* variants with a wide range of drug sensitivity scores were selected for validation. Cells were incubated with 3 mM 6TG for 24 hr, and then subjected to DNA-TG quantification as described previously. P value was estimated using the Spearman test.

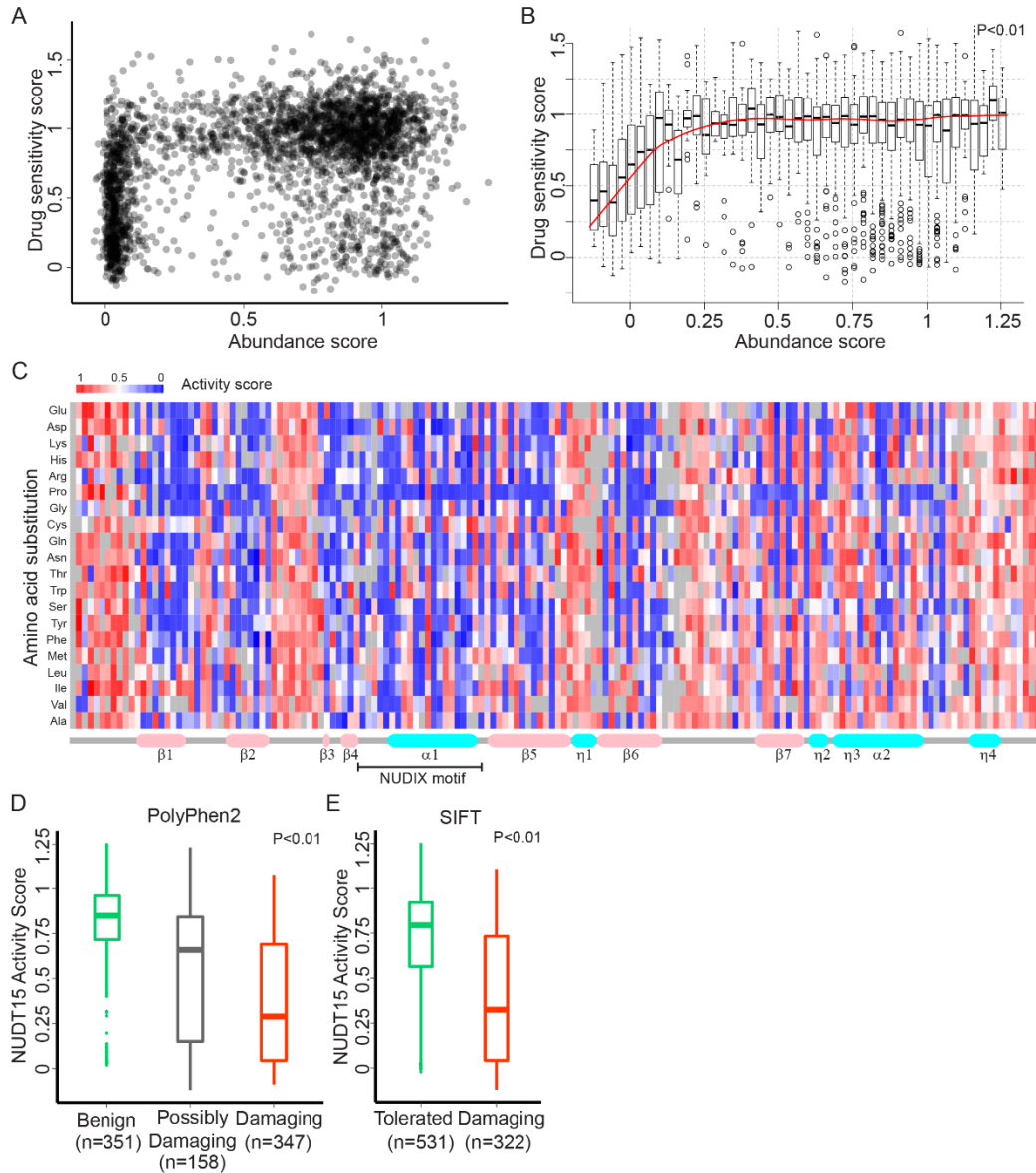


Fig. S6. The development of NUDT15 activity score. **A)** Relationship between abundance score and drug sensitivity score. **B)** Correlation of abundance score and drug sensitivity score across *NUDT15* variant. P value was estimated by using the correlation test. **C)** Heatmap of NUDT15 activity score across 2,844 variants. Each column represents an amino acid residue in NUDT15 protein [from 1 to 164], and 19 rows indicate all 19 possible missense changes from the wildtype at this position. Red and blue denotes wildtype-like and damaging NUDT15 variants, respectively. Secondary structures are schematically indicated below the heatmap. Correlation between NUDT15 activity score with PolyPhen2 (**D**) and SIFT (**E**) predictions. Numbers represent variants classified in each category. P values were estimated by using the Mann–Whitney–Wilcoxon test. P value was estimated by using the correlation test.

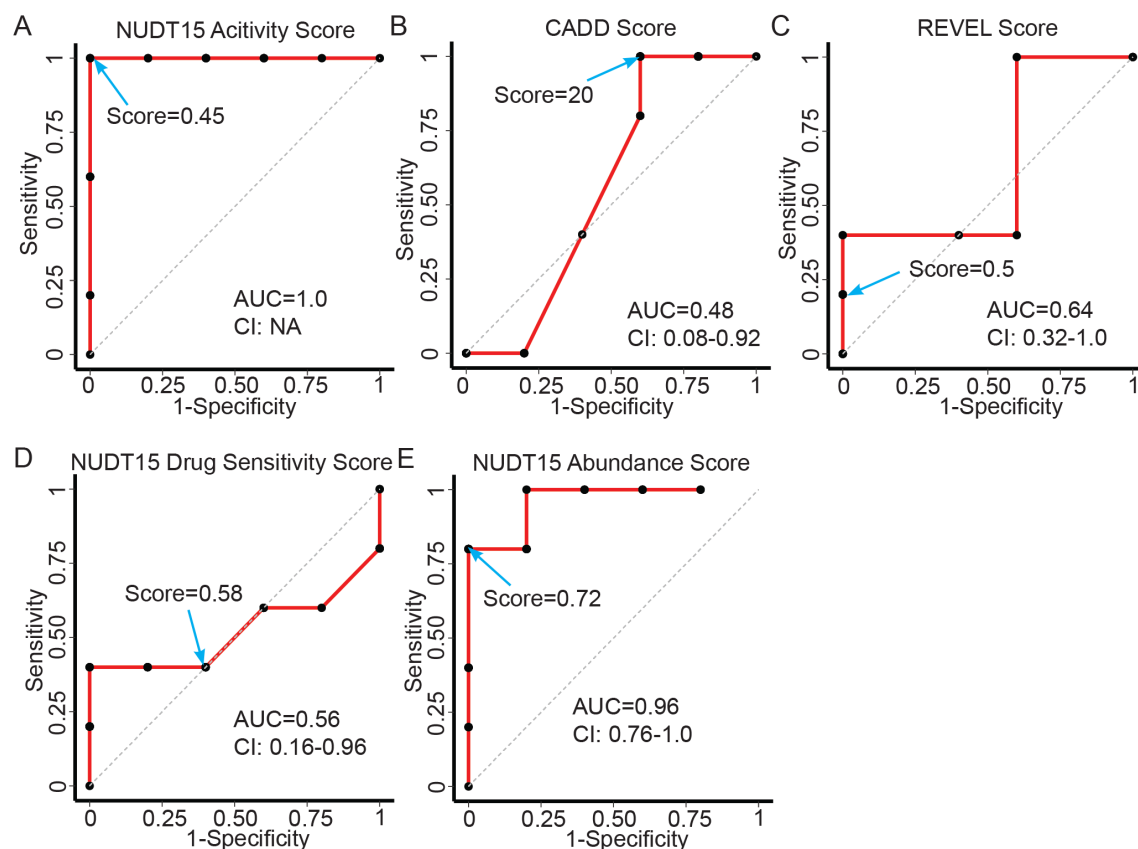


Fig. S7. Receiver operating characteristic curves (ROC) analysis of *NUDT15* risk variant prediction. ROC was plotted for each model/scoring system: A) NUDT15 activity score; B) CADD score; C) REVEL score; D) NUDT15 drug sensitivity score; E) NUDT15 abundance score. CI: 95 confidence interval of AUC. Proposed cutoff for each score is indicating by arrow. Analysis was based on 10 variants identified in 2,839 patients treated with thiopurine and NUDT15 pharmacogenetic variants were defined on the basis of clinical toxicity experienced by individuals who carried these variants (see Supplementary Text).

Table S1. . Clinically observed *NUDT15* variants and their impact on thiopurine toxicity. 10 missense variants were identified by sequencing 2,398 patients, and were defined as either “toxicity-related” or “non-risk” (See Methods). For common variants, P values were estimated by using the linear regression model after adjusting for cohorts. Tolerated MP dosage in WT controls was either described in our previous publications *Nat Genet* 48:367 and *J Clin Oncol* 33:1235 (*), or estimated from ancestry-matched WT individuals treated on the same ALL protocol (†), with details included in Supplementary text. Abbreviations. NA, not applicable; sd, standard deviation; UK, United Kingdom; USA, United States of America; WT, wildtype.

Variant	Cohort(s)	Case Number	Tolerated MP dosage (mg/m ²), mean (± sd)		Frequency	P value	Co-efficient	Final class
			Case	Ancestry-matched WT control				
Arg139Cys	USA	27	50.5 (± 18.3)	63.5 (± 14.8)*	Common	4.0 × 10 ⁻⁹	-11.2	Toxicity-related
	Singapore	21	32.7 (± 12.7)	44.4 (± 13.3)*				
	Japan	15	32.8 (± 21.8)	45.9 (± 17.9)*				
	Guatemala	13	29.3 (± 18.0)	34.2 (± 14.4)*				
Arg139His	USA	5	52.6 (± 21.7)	65.0 (± 13.6)*	Common	0.029	-9.5	Toxicity-related
	Guatemala	6	27.1 (± 19.9)	34.2 (± 14.4)*				
Val18Ile	USA	2	43.5 (± 5.5)	60.0 (± 16.9)*	Common	0.36	-6.6	Non-risk
	Singapore	2	42.0 (± 11.2)	44.4 (± 13.3)*				
	Japan	1	50.4	45.9 (± 17.9)*				
Gln6Glu	USA	1	58.3	60.0 (± 16.9)*	Rare	NA	NA	Non-risk
Arg11Gln	Hong Kong	1	50.0	45.9 (± 14.9)†	Rare	NA	NA	Non-risk
Lys33Glu	UK	1	NA	NA	Rare	NA	NA	Toxicity-related
Arg34Thr	Singapore	1	17.9	44.4 (± 13.3)*	Rare	NA	NA	Toxicity-related
	Taiwan	1	16.4	45.1 (± 15.3)†				

Val75Gly	UK	1	NA	NA	Rare	NA	NA	Toxicity-related
Ser83Tyr	India	1	52.9	49.7 ($\pm$ 3.5) [†]	Rare	NA	NA	Non-risk
Val93Ile	India	1	50.5	49.7 ($\pm$ 3.5) [†]	Rare	NA	NA	Non-risk

Table S2. Primer sequences for library construction and cloning.

Sequence (5'-3')	
<u>attB-NUDT15-GFP-IRES-mCherry construction</u>	
Fragment1_F	ATAGGGCGAATTGGGTACCG
Fragment1_R	GAGCTCTTGCGGTGATATCTGCGGATGATCCTGA
Fragment2_F	ATCACCGCAAGAGCTCCACGCCACCATGACGGCCA
Fragment2_R	CTAGCGCATGCTCCGAGATGATTCC
Fragment3_F	CTCGGAGCATGCGCTAGCGGAGTGAGCAAG
Fragment3_R	GAATTCCTTAAGTTACTACTTGACAGCTCGTCC
Fragment4_F	GTAGTAACTTAAGAATTCACC
Fragment4_R	AGATCAGATCCCATAACAATGG
<u>Barcoding library</u>	
Barcode_F	CTTAAGAATTCACCGGTCTGACCTCCTTCTCCCTCTCTTCAGGTCTGCAATTGCGTGAGTAGGGTCGACCAAGAACCGCTAGATGCGTCGCTGTACAAATAGTTNNNNNNNNNNNNNNNNNTGCGAGTAGTCGTTTCTTCGAACCGTCGACGGTACCGCGGGCCCGGGATCCAC TAGTTCTAGATAACTGATC
Barcode_R	GATCAGTTATCTAGAACTAGTGGAT
<u>Library preparation for Pacific Biosciences SMRT sequencing</u>	
XbaI_SMRTBell	/5Phos/CTAGCTCTCTCTTTTCTCTCCTCCTCCGTTGTTGTTGTTGAGAGAGAG
SacI_SMRTBell	/5Phos/ATCTCTCTCTTTTCTCTCCTCCTCCGTTGTTGTTGTTGAGAGAGATAGCT
<u>Amplification for library sequencing</u>	
Seq_1 st _F	GAGAACGTATGTCGAGGTAGGC
Seq_1 st _R	GGGTTAGCAAGTGGCAGCCTGATCAGTTATCTAGAACTAGTGGA
Seq_2 nd _F	AATGATACGGCGACCAACCGAGATCTACACGAATTCACCGGTCTGACCTC
Seq_2 nd _R1	CAAGCAGAAGACGGGCATACGAGATGATGTACAGGGTTAGCAAGTGGCAGCCT
Seq_2 nd _R2	CAAGCAGAAGACGGGCATACGAGATTGCTTTGGGGTTAGCAAGTGGCAGCCT
Seq_2 nd _R3	CAAGCAGAAGACGGGCATACGAGATTACAATACGGGTTAGCAAGTGGCAGCCT
Seq_2 nd _R4	CAAGCAGAAGACGGGCATACGAGATATACGTGAGGGTTAGCAAGTGGCAGCCT
Seq_2 nd _R5	CAAGCAGAAGACGGGCATACGAGATTCCAGTTGGGGTTAGCAAGTGGCAGCCT
Seq_2 nd _R6	CAAGCAGAAGACGGGCATACGAGATTTATCCTGGGGTTAGCAAGTGGCAGCCT
Seq_2 nd _R7	CAAGCAGAAGACGGGCATACGAGATGTACTCATGGGTTAGCAAGTGGCAGCCT
Seq_2 nd _R8	CAAGCAGAAGACGGGCATACGAGATCTCTATACGGGTTAGCAAGTGGCAGCCT
Seq_2 nd _R9	CAAGCAGAAGACGGGCATACGAGATATTCTCGAGGGTTAGCAAGTGGCAGCCT
Seq_2 nd _R10	CAAGCAGAAGACGGGCATACGAGATCAATCTATGGGTTAGCAAGTGGCAGCCT
Seq_2 nd _R11	CAAGCAGAAGACGGGCATACGAGATCGAGCGACGGGTTAGCAAGTGGCAGCCT
Seq_2 nd _R12	CAAGCAGAAGACGGGCATACGAGATTCGATTATGGGTTAGCAAGTGGCAGCCT
Seq_2 nd _R13	CAAGCAGAAGACGGGCATACGAGATTGGGCTACGGGTTAGCAAGTGGCAGCCT
Seq_2 nd _R14	CAAGCAGAAGACGGGCATACGAGATTCATCTCGGGTTAGCAAGTGGCAGCCT
Seq_2 nd _R15	CAAGCAGAAGACGGGCATACGAGATGTCGCTCGGGGTTAGCAAGTGGCAGCCT
Seq_2 nd _R16	CAAGCAGAAGACGGGCATACGAGATTCAGCAACGGGTTAGCAAGTGGCAGCCT
<u>Mutagenesis for selected variants</u>	
Val16Asp_F	CCAGGAGTCGGAGACGGAGTCGTGGTG
Val16Asp_R	CACCACGACTCCGTCTCCGACTCCTGG
Val18Ala_F	GTCGGAGTCGGAGCCGTGGTGACCAGC
Val18Ala_R	GCTGGTCACCACGGCTCCGACTCCGAC
Leu30Val_F	CCGCGTTGCGTCGTCCTGGGGAAGA
Leu30Val_R	TCTTCCCAGGACGACGCAACGCGG

Lys33Glu_F	CGTCCTCCTGGGGGAGAGGAAAGGCTC
Lys33Glu_R	GAGCCTTTCTCTCCCCAGGAGGACG
Ser37Trp_F	TGGGGAAGAGGAAAGGCTGGGTGGAGCT
Ser37Trp_R	AGCTCCAACCCAGCCTTTCTCTCCCA
Val38Ala_F	GAGGAAAGGCTCGGCTGGAGCTGGCAGTT
Val38Ala_R	AACTGCCAGCTCCAGCCGAGCCTTTCTC
Phe52Tyr_F	CCCTGGAGGTCATCTGGAGTACGGTAAACCTGGG
Phe52Try_R	CCCAGGTTTCACCGTACTCCAGATGACCTCCAGGG
Gly53Ser_F	GAGGTCATCTGGAGTTCAGTGAAACCTGGGAAGAA
Gly53Ser_R	TTCTTCCAGGTTTCACTGAACTCCAGATGACCTC
Val75Gly_F	GCTCTTCACCTGAAAAATGGTCACTTTGCCTCAGTTG
Val75Gly_R	CAACTGAGGCAAAGTGACCATTTTTTCAGGTGAAGAGC
Val93Ile_F	TCTTTCATTGAGAAGGAGAATTACCATTATATTACTATATTAATGAAAGGAGAAG
Val93Ile_R	CTTCTCCTTTCATTAATATAGTAATATAATGGTAATTCTCCTTCTCAATGAAAGA
Met97Thr_F	AGGAGAATTACCATTATGTTACTATATTAACGAAAGGAGAAGTGGATG
Met97Thr_R	CATCCACTTCTCCTTTCGTTAATATAGTAACATAATGGTAATTCTCCT
Glu108Ala_F	GATGTGACTCATGATTCAGCACCAAGAATGTAGAGCCT
Glu108Ala_R	AGGCTCTACATTCTTTGGTGCTGAATCATGAGTCACATC
Lys110Thr_F	GTGACTCATGATTCAGAACCAACGAATGTAGAGCCTGAAAAAAT
Lys110Thr_R	ATTTTTTTCAGGCTCTACATTCGTTGGTTCTGAATCATGAGTCAC
Arg139Cys_F	CAGCTTTTCTGGGGACTGTGTTGTTTAAAGAACAAG
Arg139Cys_R	CCTTGTTCTTTTAAACAACACAGTCCCCAGAAAAGCTG

Table S3. Protocol-defined standard MP dosage and criteria for dose adjustment. ALL protocols in this study have a planned MP dosage ranging from 50 to 75 mg/m² during maintenance therapy. MP dosage is clinically adjusted according to a target range of WBC and/or ANC. Abbreviations. ANC, absolute neutrophil count; HR, high risk; IR, intermediate risk; MP, 6-mercaptopurine; SR, standard risk; VHR, very-high risk; WBC, white blood cell; USA, United States of America.

Cohort	Planned MP dosage (mg/m²)	Criteria for dose adjustment
USA	75	ANC 0.75 – 1.5 x 10 ⁹ /l
Singapore	50 (SR, IR), 75 (HR)	WBC 2.0 – 4.0 x 10 ⁹ /l
Japan	50	WBC 2.0 – 3.0 x 10 ⁹ /l
Guatemala	50 - 75	WBC 1.5 – 3.0 x 10 ⁹ /l
Taiwan	60 (SR, IR), 75 (VHR)	WBC 1.8 – 3.0 x 10 ⁹ /l ANC 0.5 – 1.2 x 10 ⁹ /l
Hong Kong	50	WBC 1.8 – 3.0 x 10 ⁹ /l ANC 0.5 – 1.2 x 10 ⁹ /l
India	60	ANC 0.75 – 1.5 x 10 ⁹ /l

Table S4. Sensitivity, specificity, and AUC for different prediction models of *NUDT15* risk variants. CI indicates 95% confidence interval. CI for sensitivity and specificity was calculated using Wilson method. CI for AUC was calculated by pROC package based on “bootstrap”.

Score	Sensitivity (CI)	Specificity (CI)	AUC (CI)	Cutoff*
NUDT15 activity score	1 (0.57-1)	1 (0.57-1)	1 (NA)	0.45
NUDT15 drug sensitivity score	0.2 (0.04-0.62)	1 (0.57-1)	0.56 (0.16-0.96)	0.58
NUDT15 abundance score	0.8 (0.38-0.96)	1 (0.57-1)	0.96 (0.76-1)	0.72
CADD	1 (0.57-1)	0.4 (0.12-0.77)	0.48 (0.08-0.92)	20
REVEL	0.2 (0.04-0.624)	1 (0.57-1)	0.64 (0.32-1.0)	0.5

Table S5. List of four variants that showed effects on HEK293T cell growth. Library-transfected cells were cultured for 2 or 6 days without thiopurine. Genomic DNA was extracted and subjected to sequencing to quantify variant frequency in each population. Only 4 variants showed a statistical significant alteration in prevalence between these two timepoints. Adjusted p-values were estimated by Bonferroni correction for the number of variants tested.

Var	Abundance score	Drug sensitive score	Activity score	log2 FC (no drug: Day 6 vs. Day 2)	adjusted p-value (no drug: Day 6 vs. Day 2)
Val81Ala	0.74	1.24	0.74	-0.80	1.67E-04
Asn111Pro	0.73	0.25	0.25	0.84	6.58E-03
Lys150Asn	0.97	1.39	0.97	-1.08	3.96E-06
Lys87Ile	0.80	1.43	0.80	-1.01	8.84E-03

Dataset S1 (separate file). List of NUDT15 abundance score, drug sensitivity score and activity score for all variants.

References

1. Matreyek KA, *et al.* (2018) Multiplex assessment of protein variant abundance by massively parallel sequencing. *Nature genetics* 50(6):874.
2. Matreyek KA, Stephany JJ, & Fowler DM (2017) A platform for functional assessment of large variant libraries in mammalian cells. *Nucleic acids research* 45(11):e102-e102.
3. Li H & Durbin R (2010) Fast and accurate long-read alignment with Burrows-Wheeler transform. *Bioinformatics* 26(5):589-595.
4. Kircher M, *et al.* (2014) A general framework for estimating the relative pathogenicity of human genetic variants. *Nat Genet* 46(3):310-315.
5. Ioannidis NM, *et al.* (2016) REVEL: An Ensemble Method for Predicting the Pathogenicity of Rare Missense Variants. *Am J Hum Genet* 99(4):877-885.
6. Adzhubei IA, *et al.* (2010) A method and server for predicting damaging missense mutations. *Nat Methods* 7(4):248-249.
7. Ng PC & Henikoff S (2003) SIFT: Predicting amino acid changes that affect protein function. *Nucleic Acids Res* 31(13):3812-3814.
8. Moriyama T, *et al.* (2016) NUDT15 polymorphisms alter thiopurine metabolism and hematopoietic toxicity. *Nat Genet* 48(4):367-373.
9. Walker GJ, *et al.* (2019) Association of Genetic Variants in NUDT15 With Thiopurine-Induced Myelosuppression in Patients With Inflammatory Bowel Disease. *JAMA* 321(8):773-785.
10. Yang JJ, *et al.* (2015) Inherited NUDT15 Variant Is a Genetic Determinant of Mercaptopurine Intolerance in Children With Acute Lymphoblastic Leukemia. *Journal of clinical oncology : official journal of the American Society of Clinical Oncology* 33(11):1235-1242.
11. Liang DC, *et al.* (2010) Long-term results of Taiwan Pediatric Oncology Group studies 1997 and 2002 for childhood acute lymphoblastic leukemia. *Leukemia* 24(2):397-405.
12. Liang DC, *et al.* (2016) NUDT15 gene polymorphism related to mercaptopurine intolerance in Taiwan Chinese children with acute lymphoblastic leukemia. *Pharmacogenomics J* 16(6):536-539.
13. Khera S, *et al.* (2019) Prevalence of TPMT, ITPA and NUDT 15 genetic polymorphisms and their relation to 6MP toxicity in north Indian children with acute lymphoblastic leukemia. *Cancer Chemother Pharmacol* 83(2):341-348.